

END TERM EXAMINATION
THIRD SEMESTER [B. TECH] JANUARY 2024

Paper Code: CIC 211**Subject: Object Oriented Programming Using C++**

Time: 3 Hours
Maximum Marks: 75

Note: Attempt any five questions including Q. No. 1 which is compulsory. Select one question from each unit.

Q. No.	Question	Marks
1	Attempt all questions: (a) What is an inline function and why is it used in C++? (b) What is a generic function? Explain with an example. (c) What are different derived data types in C++? (d) What is a friend function and why is it used in C++? (e) Explain C++ Garbage Collection and memory management.	5 × 3 = 15
UNIT – I		
2	(a) Differentiate between call by value and call by reference with program code. (7.5 Marks) (b) What are different types of operators used in C++? Explain with examples. (7.5 Marks)	7.5 + 7.5 = 15
3	[OR] (a) Explain features of Object Oriented Programming using suitable examples. (7.5 Marks) (b) What is function overloading? Explain its usage using suitable examples. (7.5 Marks)	7.5 + 7.5 = 15
UNIT – II		
4	(a) Explain the use of constructor and destructor in a class using examples. (7.5 Marks) (b) What is dynamic memory allocation? Explain with program code. (7.5 Marks)	7.5 + 7.5 = 15
5	[OR] (a) What is operator overloading? Overload + operator to concatenate 2 strings. (7.5 Marks) (b) Explain different types of constructors using suitable examples. (7.5 Marks)	7.5 + 7.5 = 15

[P.T.O.]

Q. No.	Question	Marks
	UNIT – III	
6	(a) Define Inheritance and explain its different types using program code. (7.5 Marks) (b) What is a pure virtual function and explain when it is used with program code. (7.5 Marks)	7.5 + 7.5 = 15
7	[OR] (a) Differentiate between composition and containership with program code. (7.5 Marks) (b) What is run time polymorphism? Explain with program code. (7.5 Marks)	7.5 + 7.5 = 15
	UNIT – IV	
8	(a) Explain C++ stream classes using suitable examples. (7.5 Marks) (b) Write a program in C++ to create a generic class for stack operations. (7.5 Marks)	7.5 + 7.5 = 15
9	[OR] (a) Write a program in C++ to read content from one file and write in another. (7.5 Marks) (b) What is the Standard Template Library (STL)? Explain iterators and vectors in detail. (7.5 Marks)	7.5 + 7.5 = 15

* * * * *